

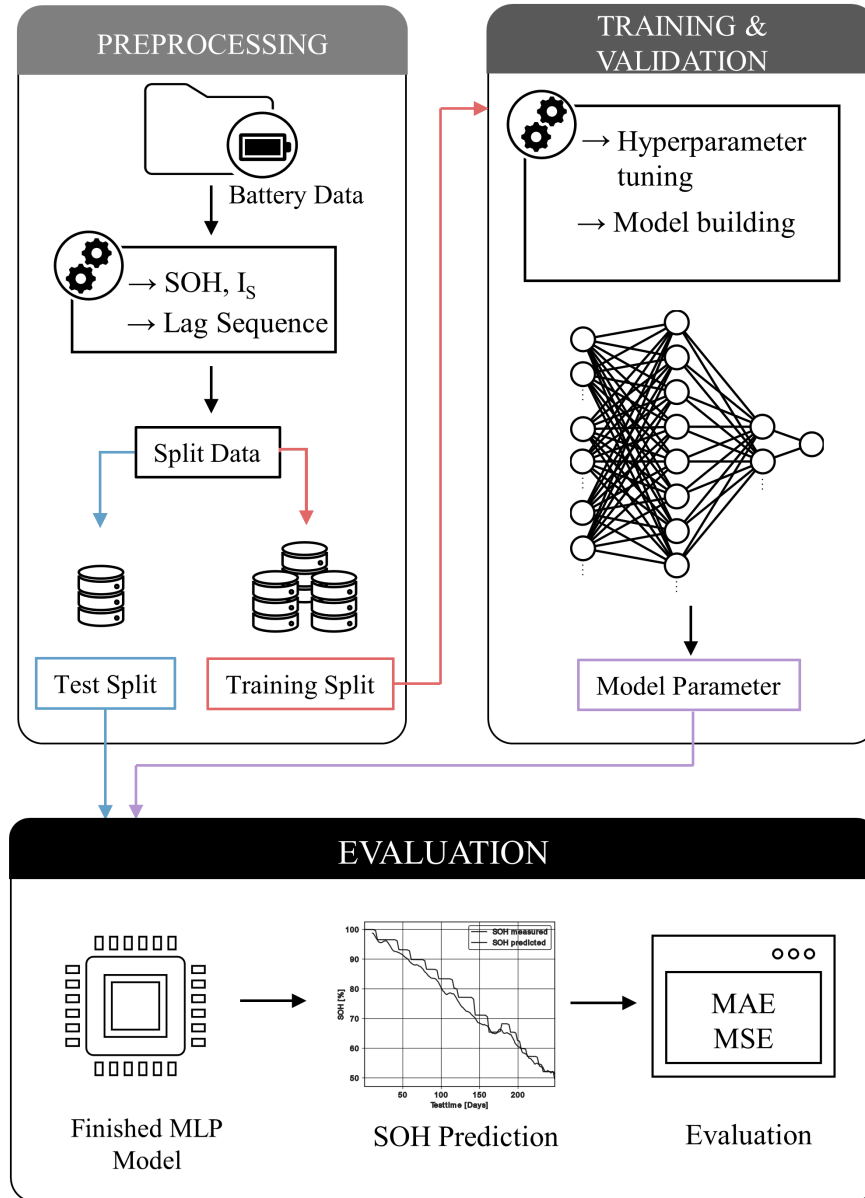


Figure 5.3: Workflow of the proposed MLP-based SOH estimation approach, from raw-cell preprocessing and data splitting through training and hyperparameter tuning to prediction and final evaluation.

5.2.5 Data Scaling

All input features are scaled to the interval between 0 and 1 before training.[108] This step is necessary because the numerical ranges of the features differ strongly: voltage remains in a narrow interval between 2,75 V and 4,2 V, whereas cumulative